

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2028

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Coursework: Data Structures, Advanced Algorithms, Machine Organization and Programming, Intro to Computer Engineering, Cryptography, Artificial Intelligence, Differential Equations, Operating Systems, Big Data, Computer Science Capstone, AP Computer Science A, Game Development, Web Development

TECHNOLOGIES

Languages: Java, Python, TypeScript, Node.js, R, Rust, JavaScript, C#, C++, SQL, Linux/Bash

Backend & Systems: Kafka, Kubernetes, Docker, MongoDB, PostgreSQL, SQLite, Redis, SQL, NoSQL, REST APIs, FastAPI, Supabase, Microservices, CI/CD, Event-Driven Architecture, Distributed Systems, System Design, Software Design, Scalable Architecture, Cloud

Frontend & Mobile: React.js, Android SDK, Spring Boot, HTML/CSS

Developer Tools: Software Engineering, Git, GitHub, GitHub Actions, Linux, Version Control, pytest, Unit Testing, Debugging, Code Review, Production Code, Agile, Cross-Functional Collaboration, Object-Oriented Design

Libraries & AI: NumPy, Pandas, PyTorch, Scikit-learn, Hugging Face, LLM APIs, RAG, OpenCV, Sentence-Transformers

EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Optimized multi-objective satellite scheduling via memoization + dynamic programming in Python, **reducing constraint violations** under real-time operational constraints
- Designed scalable Kafka event-driven pipeline to MongoDB with **at-least-once delivery, fault-tolerant streaming**, and full system decoupling for mission-critical satellite ops
- Deployed containerized Docker test suite on Kubernetes via **GitHub Actions CI/CD**; comprehensive unit testing and debugging coverage ensuring production reliability and software engineering quality across all system components

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Student Research Intern

- Built agentic scRNA-seq pipeline optimization system in Python/R on Linux, cutting GPU consumption by **300%**; collaborated with clinicians across multi-library environment
- Engineered TransferAgent module automating citation mapping and cross-paper scoring; reduced research review time by **~60%** via reproducible CI/CD pipeline
- Validated real-world impact of distributed pipeline on glioblastoma gene therapy via agile collaboration with multidisciplinary research team

Inspirat AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built **skin cancer detection pipeline** (OpenCV + scikit-learn/TensorFlow, NumPy/Pandas) achieving **93% classification accuracy**; published as **peer-reviewed paper** through Inspirat AI
- Trained Random Forest on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation

TECHNICAL PROJECTS

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built financial risk engine on **1,080-node ICD-10 comorbidity graph** across **45M insurance records** via SQL queries; object-oriented RAG pipeline + semantic search; React/Cytoscape.js frontend with voice interface
- Designed scalable system architecture separating NLP inference (Groq Llama) from deterministic financial logic (Python engine); ensures **ACID-compliant, hallucination-free** outputs — production-grade software design

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Architected failure-driven self-improvement loop across **5,000+ task benchmark**, outperforming ReAct by **15pp**; built AgentBench + LLM-as-judge eval infrastructure with GitHub Actions CI/CD

LLM-Based SAT Practice Test Generator (<https://github.com/AkashGogate/SATPracticeTestGenerator>)

Jun 2024

- Built full-stack SAT prep app (TypeScript/JavaScript/React.js + Node.js/FastAPI); integrated GPT-4o generative AI for adaptive question generation via REST API with session state, scoring, configurable difficulty; agile version control

USTA Tournament Explorer (<https://github.com/AkashGogate/myUSTA>)

Jun 2022

- Built Android app in Java consuming USTA's GraphQL API to surface local tournament data via REST API; sole developer end-to-end

Ecological Conservation Game (Runescape Parody) (<https://github.com/AkashGogate/EcologicalConservationBots>)

Jan 2024

- Engineered multi-threaded, AI-driven multi-agent C++ simulation with Raylib; autonomous bots exhibit emergent behaviors under ecological constraints with real-time game loop

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics; coached athletes across skill levels in cross-functional environment; managed end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations